

WASP-77Ab

R-1463

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-77_151218 · created 2026-08-01T11:01:42+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2457374.64057 +/- 0.00079 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1131 +/- 0.0043
Transit depth (Rp/Rs)^2	1.28 +/- 0.096 [%]
Orbital Inclination (inc)	89.62 +/- 0.91
Airmass coefficient 1 (a1)	1.1375 +/- 0.0045
Airmass coefficient 2 (a2)	-0.0935 +/- 0.0044
Scatter in the residuals of the lightcurve fit is	0.4 %
Best Comparison Star	None
Optimal Aperture	5.68
Optimal Annulus	15.04
Transit Duration (day)	0.08924 +/- 0.00061

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.1131 ± 0.0043 vs 0.1301 (deviation 3.95σ)
Duration fit vs published	0.08924 d vs 0.0908 d (deviation 2.61σ)
Mid-time O–C	-3.0 min
Depth SNR	26.3
Residual scatter	0.4 %

Light curve

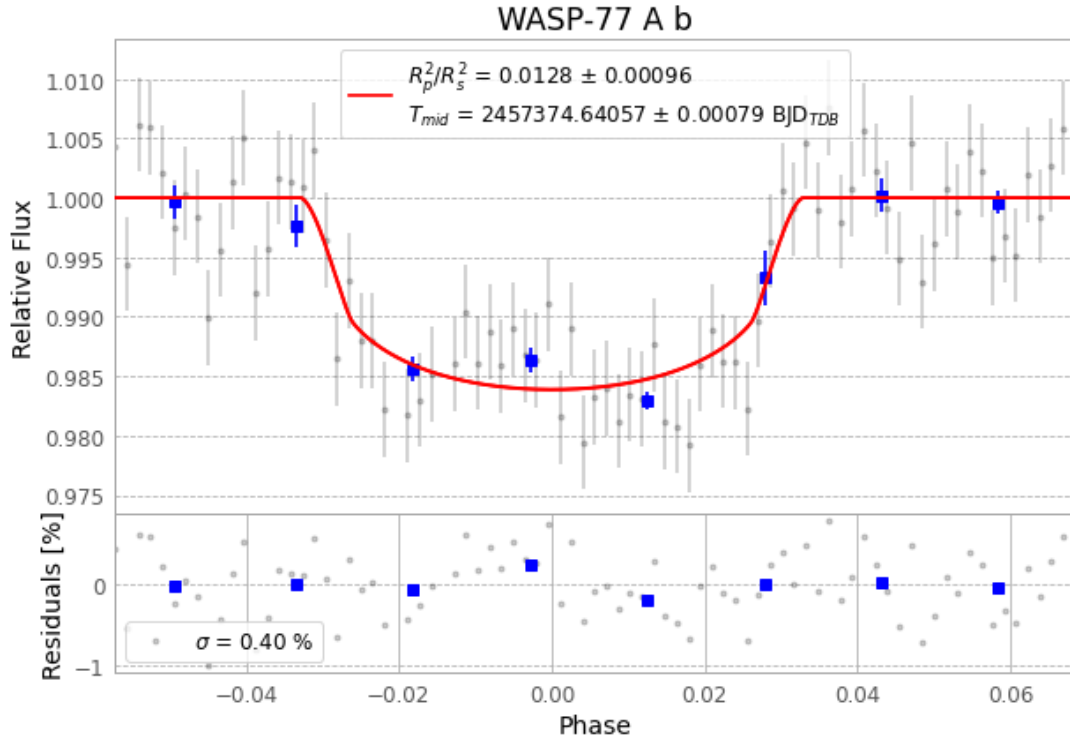


Figure 1 — FinalLightCurve_WASP-77 A b_2015-12-17.png (SHA-256 7d0f801ca39e98d5...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [124, 223], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 c47ed400813ff8fe...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

83 FITS frames (55 MB), WASP-77151218012412.FITS → WASP-77151218053012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-77-151218.log (09e1cb69d11a...), FinalLightCurve_WASP-77 A b_2015-12-17.png (7d0f801ca39e...), FinalLightCurve_WASP-77 A b_2015-12-17.csv (a1e1b3a29693...)

Compute resources

Wall time 135.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 11:01Z.